Architecture Laws for Machine Agency: A Synthesis of the Probe-Calibration Lineage

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Abstract

This note distills the program's practical architecture lessons from the first-order self, probe-calibration, world-responds, planning, and long-horizon papers. The empirical arc does not show consciousness, full agency, or general intelligence. It shows something narrower and more useful: small architectural changes repeatedly determine whether an agent's behavior is tied to the intended internal structure or to a proxy.

The recurring pattern is:

1. A behavior-level success appears.
2. A representation or control diagnostic shows the success is proxy-mediated.
3. A simple architecture change restores the intended causal structure.
4. A harder setting reveals the next proxy.

The resulting contribution is a ledger of architecture laws for machine agency: preserve vector-valued concern, use identifying interventions, recompute memory against the current model, normalize uncertainty before allocating inquiry, re-open inquiry after apparent convergence, cool at the decision layer rather than deleting the signal, split or gate heads when component identifiability stalls, bind memory to future action surfaces, and expose expected deployment transformations as local selection or training pressures.

1. Claim Boundary

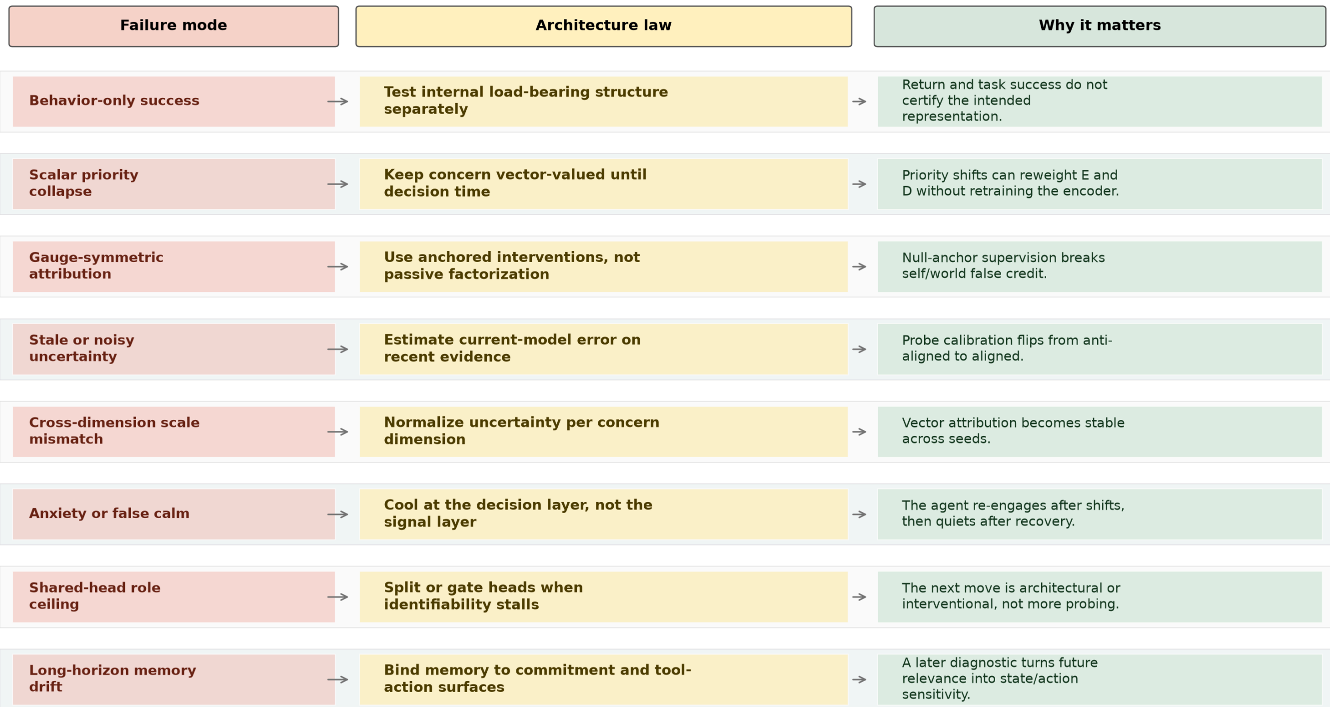
These are **minimal computational precursors**. The experiments are mostly small homeostatic bandits, plus newer long-horizon language-agent diagnostics. The right claim is not "we made conscious agents." The right claim is that the program found a sequence of cheap, testable architecture constraints that move a system from behavior-only competence toward maintained self/world, memory/action, and concern/action coupling.

The recent non-peer-reviewed preprint by Lyons, Pio-Lopez, and Levin (2026) uses the phrase **virtual governor** for a distributed signal architecture that turns global constraint violations into local incentives. That phrase is useful framing here, but not evidence. In our terms, an architecture law is a tested constraint on the signal architecture: it asks whether the local action pressure really tracks the intended global concern.

2. The Ledger

Figure 2B. Architecture laws forced by the Metric Stack

The payoff is not one clever module; it is a reusable failure → law → payoff ledger.



Failure mode	Architecture law	Agency payoff
Behavior-only success	Test internal load-bearing structure separately	Return is no longer mistaken for representation.
Scalar priority collapse	Keep concern vector-valued until decision time	Priorities can reweight E and D without retraining.
Gauge-symmetric attribution	Use anchored interventions, not passive factorization	Self/world false credit becomes measurable and correctable.
Stale or noisy uncertainty	Estimate current-model error on recent raw evidence	Probe calibration flips from anti-aligned to aligned.
Cross-dimension scale mismatch	Normalize uncertainty per concern dimension	Vector attribution becomes stable across seeds.
Learned quiet after shifts	Add a re-engagement floor or surprise detector	The agent can restart inquiry after the world changes.
Anxiety or false calm	Cool at the decision layer, not the signal layer	The agent re-engages after shifts and quiets after recovery.
Shared-head role ceiling	Split or gate heads when identifiability stalls	The next move becomes architectural or interventional.
Long-horizon memory drift	Bind memory to commitment and tool-action surfaces	Memory becomes control-relevant, not merely present.
Dispatch-surface brittleness	Harden commitment surfaces across parser, repair, and wording variants	Tool/action memory remains useful under interface stress.
Shortcut-compatible OOD success	Score or regularize structure-compatible transformations	Global deployment stress becomes local model pressure.

3. Why These Laws Matter

3.1 Memory

The current-error calibration paper gives the simplest memory rule: do not store only stale residual tokens. Store enough raw evidence to replay it through the current model. A long-horizon agent's memory is useful when it can be revalued by the policy and world model that will act now.

3.2 Planning

Planning from concern closes a loop: learned concern-shaped representation can drive a predictive policy without external reward supervision. The architecture lesson is that a planner should consume a representation that preserves the dimensions of mattering until decision time. Premature scalarization makes planning brittle under shifted priorities.

3.3 Action

The null-intervention and world-responds papers show that self/world attribution requires actions whose purpose is identification, not reward. In nonstationary worlds, the action policy also needs a way to re-open inquiry after the model becomes quiet. Learned silence is not evidence of world stability.

3.4 Reafference And Selfhood

The first-order self result is a negative law: input factorization does not identify causal source. Reafference needs a gauge-breaking signal. In this program, the gauge breaker is an active null-anchor intervention plus calibrated probe selection. This is a selfhood ingredient, not a consciousness certificate.

3.5 Long Trajectories

The long-horizon bottleneck work adds the temporal complement: a hidden clue matters only if it reaches a later commitment surface, such as a tool call, schema field, or irreversible action. Memory without future action coupling is just context. Memory with commitment coupling becomes agentic state.

The dispatch robustness follow-up sharpens the same law. The issue is not just whether a model remembers a hidden value; it is whether the value survives the surface on which it must be used: parser-facing text, aliases, generated JSON, tool repair, and black-box API dispatch wording. A sparse OpenAI GPT-4.1 Nano negative under one dispatch repair cell is therefore useful precisely because it localizes the pressure point without turning it into a broad provider claim.

3.6 Structure-Compatible Generalization

The structure-compatible generalization work adds an OOD complement. When ID evidence admits both shortcuts and transportable rules, the architecture needs a surface that asks whether the learned function preserves the transformation expected to generate deployment cases. Phase one shows this as model selection; phase two weakens the oracle by inferring supported modular shifts and using them as a compatibility regularizer. In virtual-governor terms, the training loop converts a global deployment constraint into local pressure on the model, but the claim remains finite-domain and task-family bounded.

4. Simple Changes That Moved Outcomes

The strongest "small changes, large effects" observed so far are:

1. **Current replay:** keep a compact raw buffer and recompute residuals against the present model. This changed V_probe from anti-calibrated to positively calibrated.
2. **Per-dimension normalization:** scale-normalize probe targets or thresholds. This removed the vector seed-collapse failure.
3. **Active null anchoring:** add an identifying intervention instead of relying on passive self/world factorization. This broke the attribution gauge.
4. **Re-engagement floor / surprise trigger:** add a path back into inquiry after convergence. This fixes self-confirming

silence.
5. **Decision-layer cooling:** damp the action tendency after recent probing without erasing the surprise signal. This avoids both anxiety and false calm.
6. **Commitment-surface memory:** test whether early information causally reaches a later action surface. This relocates long-horizon agency from context length to state/action coupling.
7. **Dispatch-surface hardening:** test the same commitment under aliases, parser-facing wording, generated JSON, repair, and API dispatch variants. This prevents hidden memory from evaporating at the action interface.
8. **Compatibility selection/regularization:** score or train against the transformations that define the deployment shift. This converts an OOD constraint into a local architecture pressure.

5. Major Contribution Opportunity

The major contribution is not a single module. It is a transferable method:

> For every proposed agent capability, find the proxy that lets behavior pass without the intended structure, then add the smallest architecture change that makes the structure load-bearing.

This is where the work should push hardest next. The homeostatic arc produced the laws. The next contribution is to test whether they transfer to:

- long-horizon tool agents,
- structure-compatible OOD generalization,
- multi-agent or decentralized coordination,
- learned intervention discovery rather than hand-coded null actions,
- richer memory systems whose retrieval is coupled to future commitments.

6. Critical Caveats

- The homeostatic experiments use simulator-defined viability variables and a privileged null action.
- Most results use three seeds.
- The virtual-governor preprint is useful framing but not peer-reviewed evidence.
- The laws are empirically grounded within this repo's tasks; transfer to LLM agents, robotics, markets, or biological systems remains a hypothesis.
- Consciousness language must remain bounded. These laws concern control, attribution, memory, and inquiry surfaces, not subjective experience.

References

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